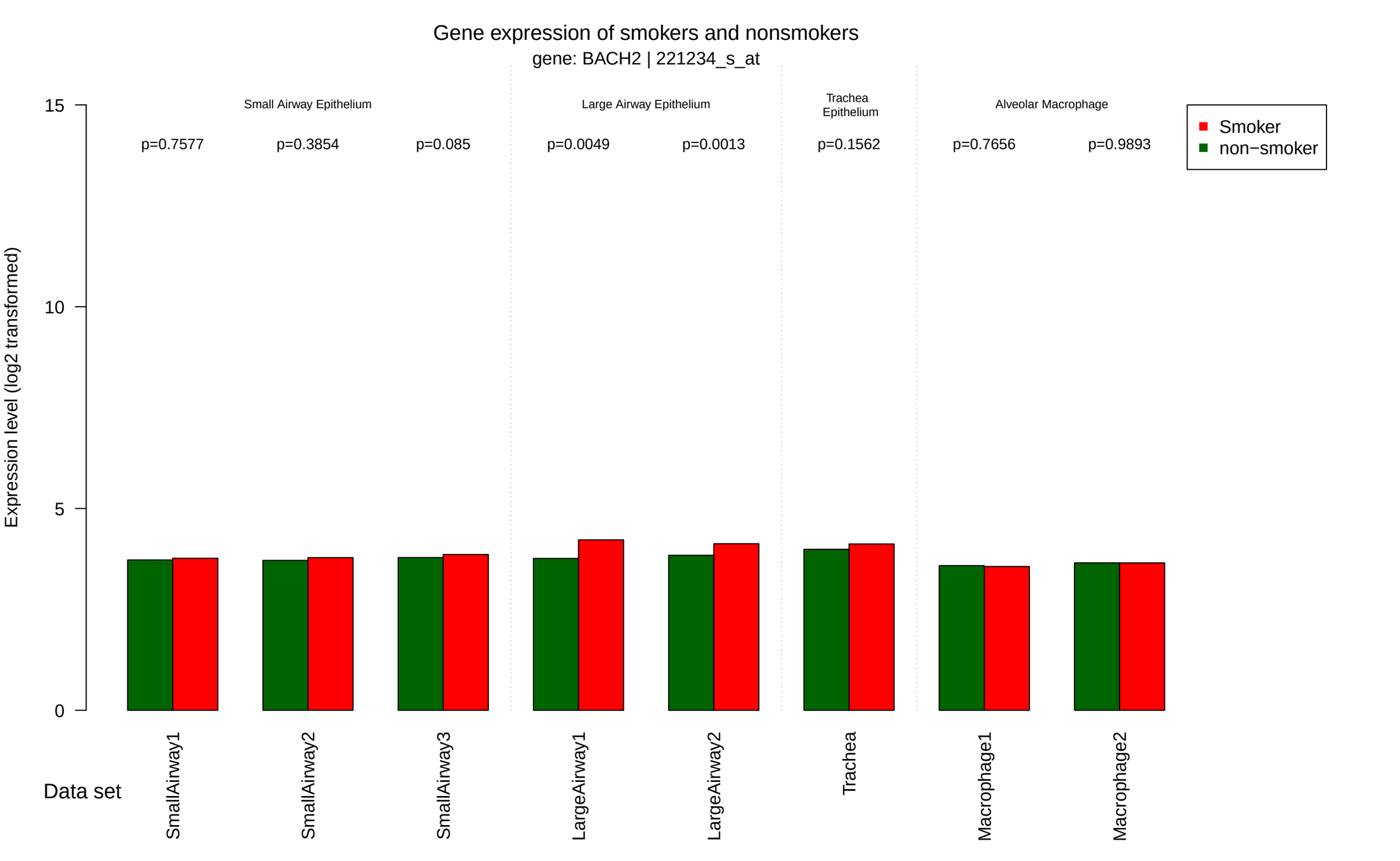
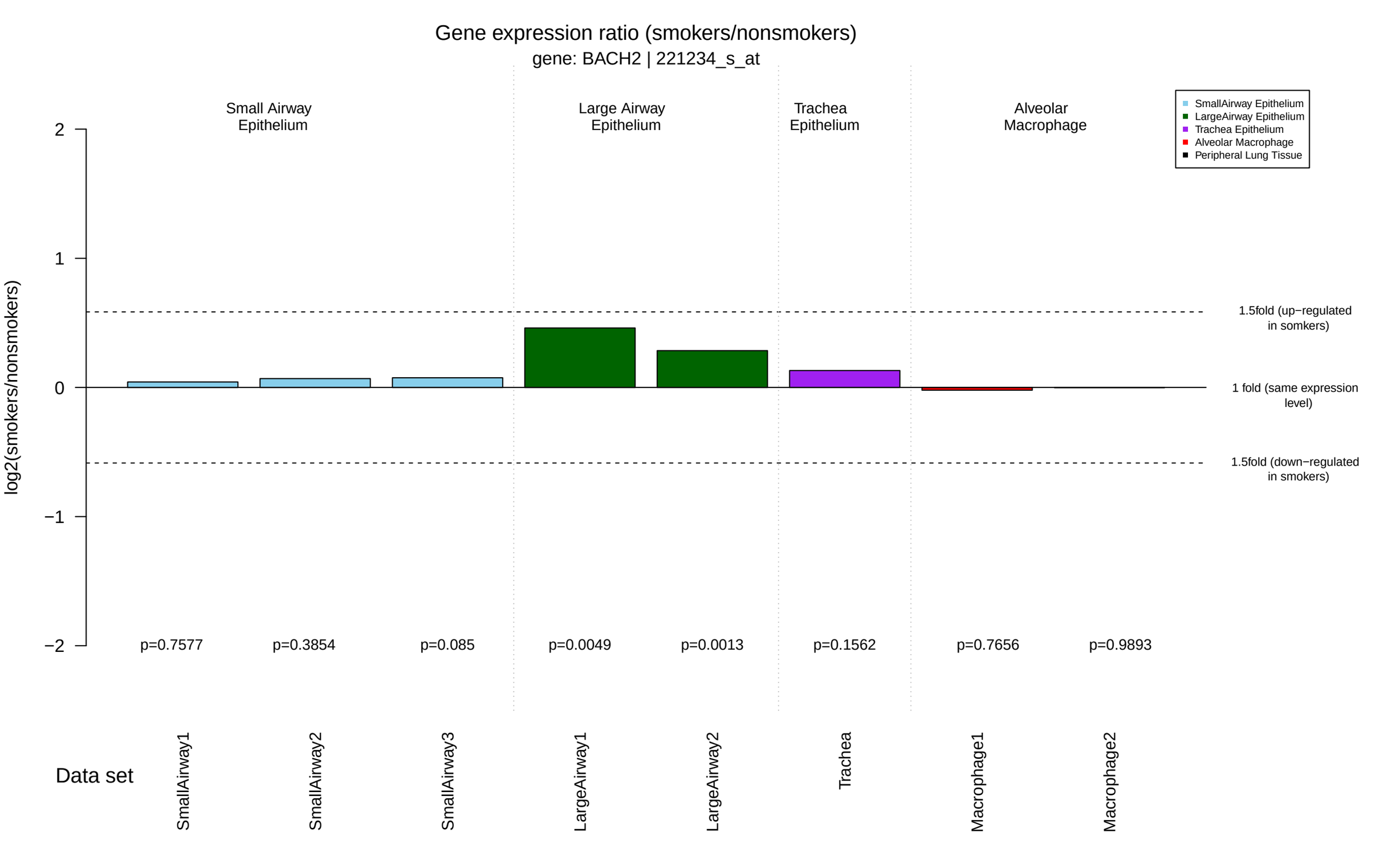
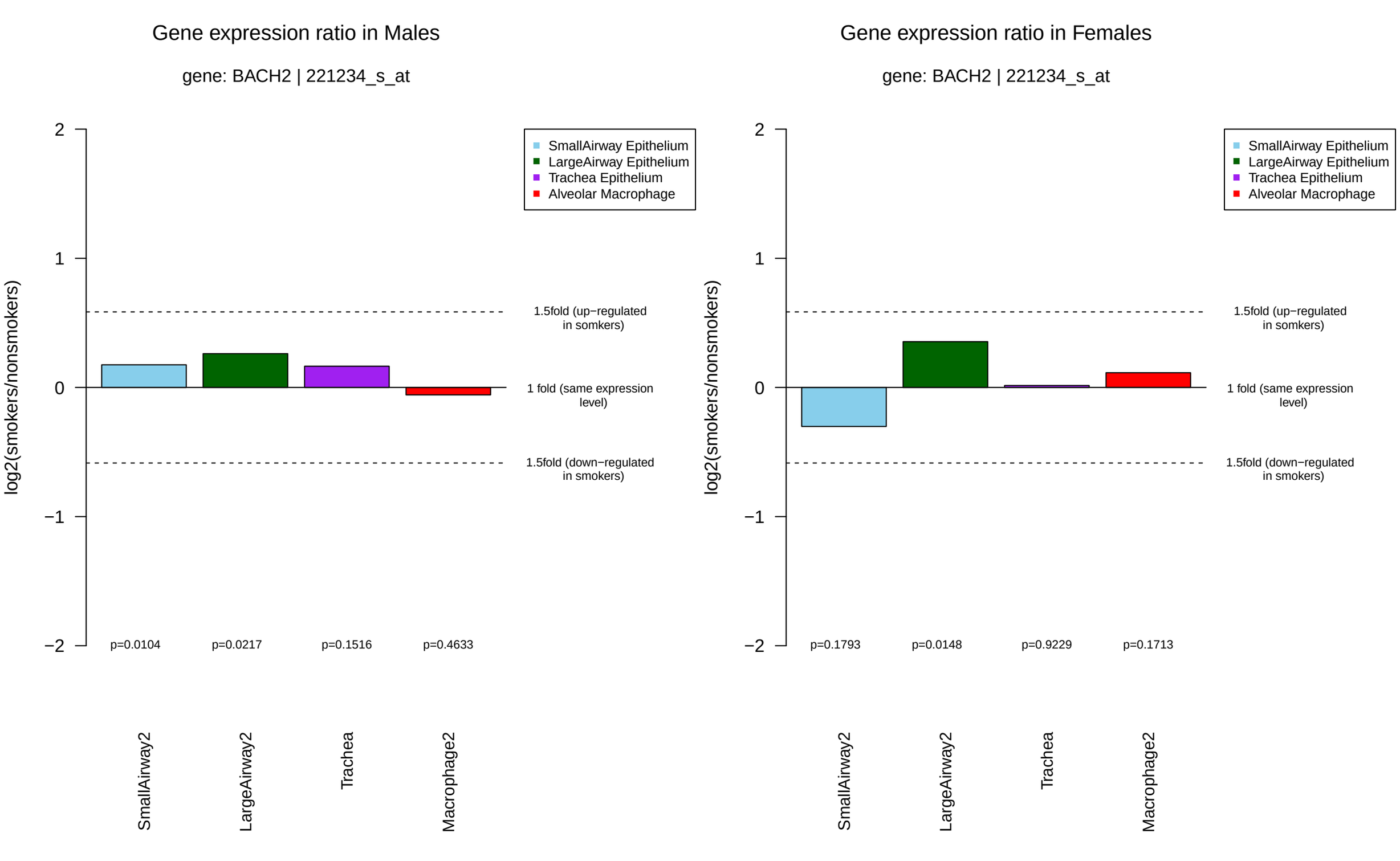
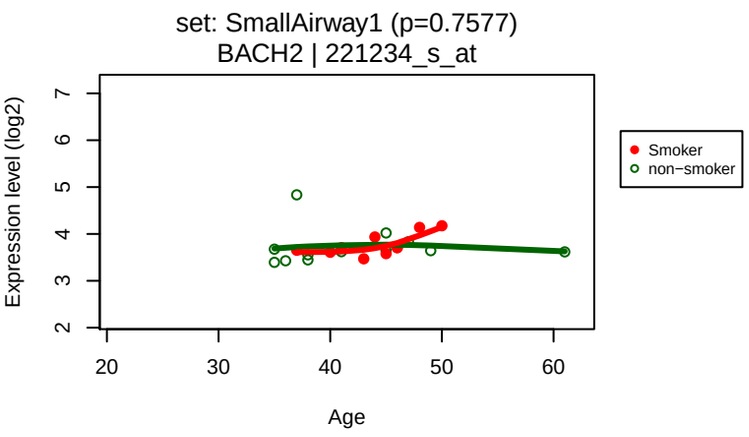
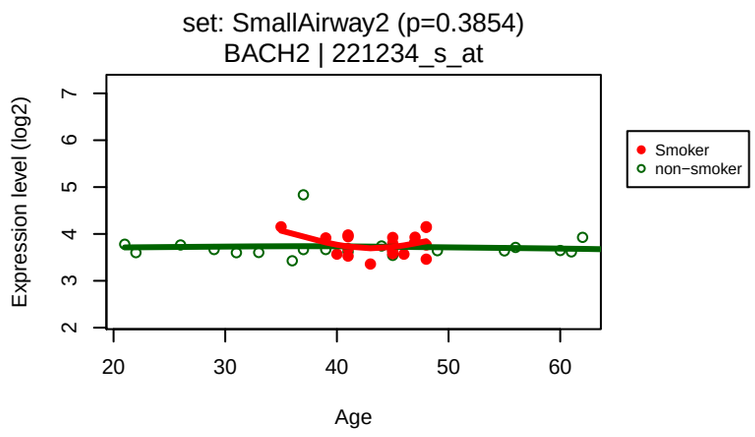
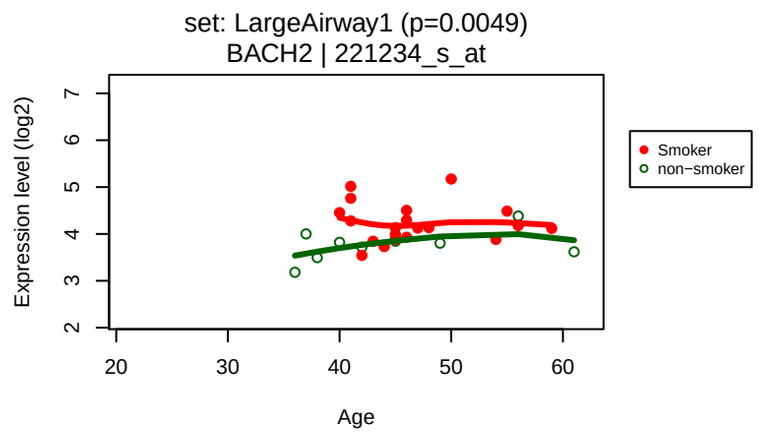

Gene expression of smokers and nonsmokers
gene: BACH2 | 221234_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7577
p=0.3854
p=0.085
p=0.0049
p=0.0013
p=0.1562
p=0.7656
p=0.9893
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

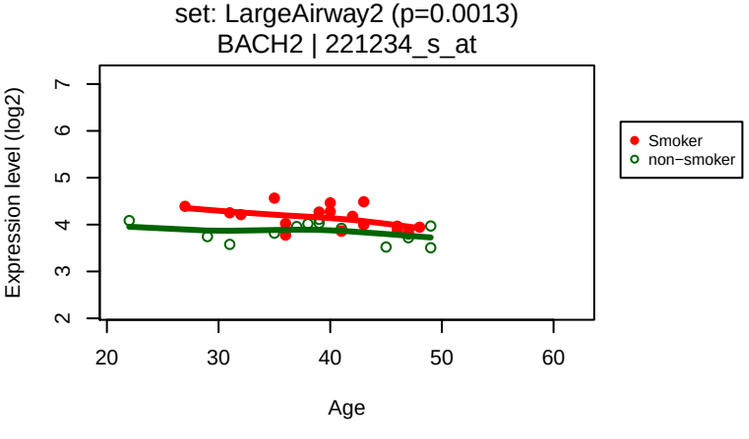

Gene expression ratio (smokers/nonsmokers)
gene: BACH2 | 221234_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7577
p=0.3854
p=0.085
p=0.0049
p=0.0013
p=0.1562
p=0.7656
p=0.9893
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

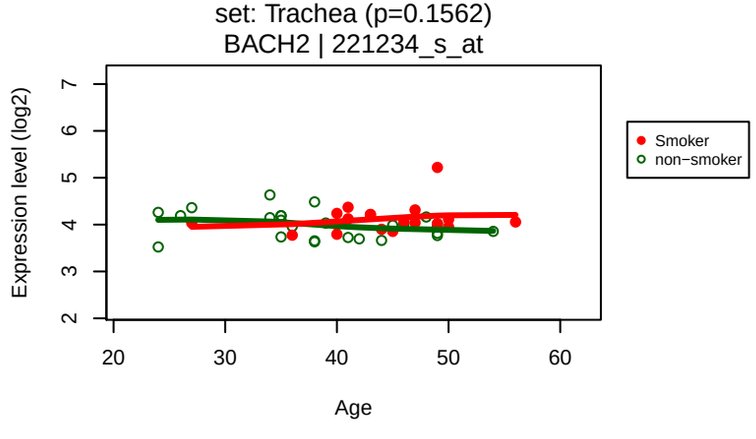

Gene expression ratio in Males
gene: BACH2 | 221234_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0104
p=0.0217
p=0.1516
p=0.4633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACH2 | 221234_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1793
p=0.0148
p=0.9229
p=0.1713
SmallAirway2
LargeAirway2
Trachea
Macrophage2

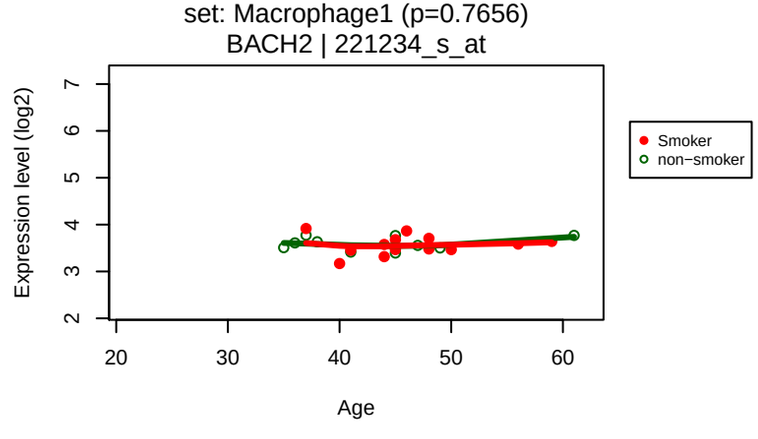

set: SmallAirway1 (p=0.7577)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker

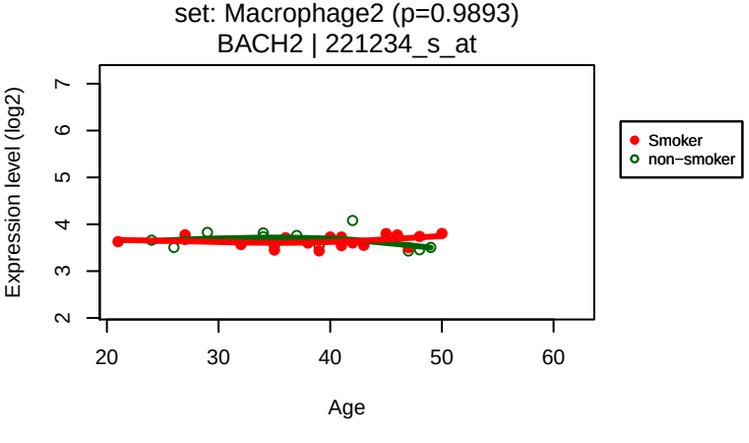

set: SmallAirway2 (p=0.3854)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker

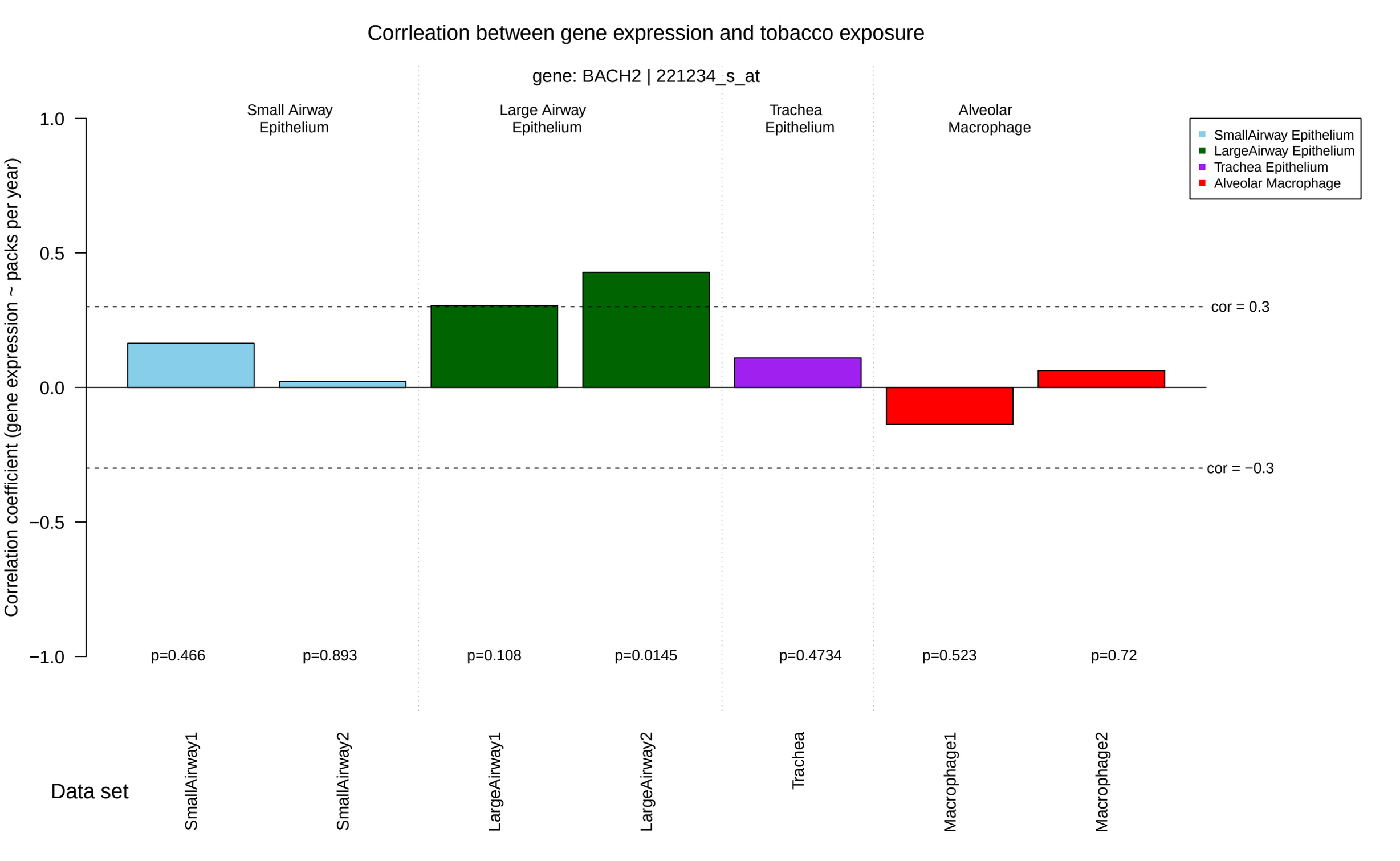

set: LargeAirway1 (p=0.0049)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0013)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1562)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7656)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9893)
BACH2 | 221234_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACH2 | 221234_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.466
p=0.893
p=0.108
p=0.0145
p=0.4734
p=0.523
p=0.72
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2